

Lecture 11: ARIMA, SARIMA, ARFIMA models

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1 ARIMA models

Let $\{Y_j, j \geq 0\}$ be a random walk. This is one of the first processes we studied at the beginning of the course. It is defined as

$$Y_j = Y_{j-1} + Z_j,$$

where $Z \sim WN(0, \sigma^2)$ and $Y_0 = Z_0 = 0$. As we already know, this is not a stationary process because

$$\mathbb{V}(Y_j) = \mathbb{V}(Y_{j-1}) + \sigma^2$$

and so,

$$\mathbb{V}(Y_j) = j \sigma^2.$$

But note that if we compute its first differences we obtain

$$X_j := Y_j - Y_{j-1} = Z_j, \quad \forall j \geq 1$$

the process of first differences, which is stationary.

This illustrates that differencing may convert non-stationary series in stationary ones. This is the basic method to extend the ARMA models to the family of ARIMA models, which include non-stationary models

Definition 1.1 *X is an ARIMA(p,d,q) model with $p, d, q \in \mathbb{N}$ if*

$$Y_j = (\text{Id} - B)^d X_j, \quad j \in \mathbb{Z}.$$

is an ARMA(p,q) model.

Process Y is process X differentiated d times and process X is process Y integrated d times. Integrated processes are called ARIMA processes. Letter I means *integrated*. Therefore, a process X is an ARIMA process if it satisfies

$$\Phi_p(B) (\text{Id} - B)^d X_j = \Theta_q(B) Z_j, \quad j \in \mathbb{Z}.$$

Obviously, ARIMA(p,0,q) means ARMA(p,q). The random walk is an ARIMA(0,1,0) process. Of course, an ARIMA process can have a trend. In this case we have

$$\Phi_p(B) (\text{Id} - B)^d X_j = \delta + \Theta_q(B) Z_j, \quad j \in \mathbb{Z}.$$

Indeed, note that for the particular case of $p = q = 0$ and $d = 1$ we have,

$$X_j = X_{j-1} + \delta + Z_j.$$

And therefore,

$$E(X_j) = E(X_{j-1}) + \delta = \dots = j \delta.$$

2 Prices

An interesting example of ARIMA process is the following. Let $S = \{S_j \mid j \geq 0\}$ be a price process in an open market, that is, under the offer and demand law. As a first attempt to model prices it is reasonable to assume

$$\text{Law}\left(\frac{S_j - S_{j-1}}{S_{j-1}}\right) = \text{Law}\left(\frac{S_{j+1} - S_j}{S_j}\right),$$

that is, relative increments are stationary, or in other words, the same law determine the relative price variation of today and the relative price variation of tomorrow. Then,

$$\text{Law}\left(\frac{S_j}{S_{j-1}}\right) = \text{Law}\left(\frac{S_{j+1}}{S_j}\right),$$

and therefore,

$$\text{Law}(\log S_j - \log S_{j-1}) = \text{Law}(\log S_{j+1} - \log S_j),$$

that is, $X_j := \log S_j - \log S_{j-1}$ is a stationary process.

Therefore, if X is an ARMA(p,q) process, then $\log S$ is an ARIMA(p,1,q) process, and it is also a logarithmic transformation of the time series of prices.

Note that if $|x| < 1$ we have the Taylor expansion

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$$

and so, writing $y = 1+x$, if $|y-1| < 1$, we have

$$\log y \simeq y - 1,$$

$$\log \frac{S_j}{S_{j-1}} \simeq \frac{S_j}{S_{j-1}} - 1 = \frac{S_j - S_{j-1}}{S_{j-1}}$$

and

$$\left| \frac{S_j}{S_{j-1}} - 1 \right| < 1 \iff \left| \frac{S_j - S_{j-1}}{S_{j-1}} \right| < 1.$$

Recall that if $\left| \frac{S_j - S_{j-1}}{S_{j-1}} \right| < 0.2$, typically, markets close the trade of the asset.

Therefore, except on critical situations, increments of logarithmic prices coincide with relative increments of prices, that is, with returns. This is the reason why the process of logarithmic prices is of main importance in Finance.

3 Dickey and Fuller test of unitary roots

A typical problem in Time Series Analysis is to distinguish between a random walk and an AR(1) process with a parameter ϕ close to 1. Recall that the two processes are very different because one is stationary and the other not. More formally, the problem is the following: given

$$X_j = \phi X_{j-1} + Z_j, \quad |\phi| \leq 1,$$

how to distinguish $\phi = 1$ from $\phi < 1$?

We can rewrite the model as

$$X_j - X_{j-1} = (\phi - 1)X_{j-1} + Z_j,$$

that is, if $\rho := \phi - 1$,

$$(Id - B)X_j = \rho X_{j-1} + Z$$

and we have to distinguish between the null hypothesis $\rho = 0$ and the alternative hypothesis $\rho < 0$.

This can be done by the statistic of Dickey-Fuller that under the null hypothesis has the expression

$$\hat{\rho} = DF_n := n \frac{\sum_{i=1}^n x_{i-1}(x_i - x_{i-1})}{\sum_{i=1}^n x_{i-1}^2}$$

and can be deduced immediately from the least squares estimator of ϕ .

Note that DF_n is less than 0, and near to 0 if the null hypothesis is true and far from it if the true hypothesis is the alternative. With a 5% of error, if DF_n is less than -1.94 we accept the alternative hypothesis that it is an AR(1) and if it is greater than -1.94 , we accept null hypothesis that it is a random walk.

An alternative to Dickey-Fuller test, that runs similarly, is the Phillips-Perron test. The Dickey-Fuller test can be generalized allowing the error Z have a mean or a deterministic trend. Or being a stationary ARMA process. In this last case, the generalization is the so called Augmented Dickey-Fuller test, and it runs similarly.

4 SARIMA models

Let $\{X_j, j \geq 1\}$ be a time series. Recall that we can write it as $X_j = X_{i+(r-1)s}$ with $i = 1, \dots, s$ and $r \geq 1$, where s is the period. For example, given a series of monthly data, $s = 12$ and r indicates the year. Therefore, the first figure is $i = 1$ and $r = 1$, that is, the month 1 of the year 1. The thirteen figure is $i = 1$ and $r = 2$, that is, $j = 1 + (2 - 1) \times 12 = 13$, and so on.

4.1 Pure seasonal models

To fix ideas assume a yearly seasonal period. A pure seasonal model assumes that there is correlation only between data of the same month, and data of different month are uncorrelated. So, in fact, we have 12 time series, one per month, uncorrelated between them.

If the model is ARMA(a, c) this means

$$X_{i+(r-1)s} - \alpha_1 X_{i+(r-2)s} - \dots - \alpha_a X_{i+(r-a)s} = U_{i+(r-1)s} - \beta_1 U_{i+(r-2)s} - \dots - \beta_c U_{i+(r-c)s}$$

with $\{U_{i+(r-1)s}, r \geq 1, i = 1, \dots, s\} \sim WN(0, \sigma^2)$. This model is denoted by $ARMA(a, c)_s \times WN(0, \sigma^2)$.

For example, a $AR(1)_{12} \times WN(0, \sigma^2)$ satisfies

$$X_{i+(r-1)12} - \alpha X_{i+(r-2)12} = U_{i+(r-1)12}, \quad |\alpha| < 1,$$

or equivalently,

$$X_j - \alpha X_{j-12} = U_j, \quad j \in \mathbb{Z}, \quad |\alpha| < 1.$$

Analogously to the non seasonal $AR(1)$ case we have

$$\gamma(0) = \frac{\sigma^2}{1 - \alpha^2}$$

and

$$\rho(j) = \begin{cases} 1, & j = 0 \\ \alpha^l, & j = l \cdot s \\ 0, & \text{otherwise} \end{cases}$$

Note that we can write

$$(\text{Id} - \alpha B^{12}) X_j = U_j, \quad j \in \mathbb{Z}$$

or

$$X_j = (\text{Id} - \alpha B^{12})^{-1} U_j, \quad j \in \mathbb{Z}.$$

4.2 Seasonal models

Think now on the model $MA(1) \times MA(1)_{12}$. Its equation is

$$X_j = (\text{Id} - \theta B)(\text{Id} - \vartheta B^{12})Z_j, \quad j \in \mathbb{Z}$$

with

$$Z \sim WN(0, \sigma^2),$$

that is, in a displayed form we have

$$X_j = Z_j - \vartheta Z_{j-12} - \theta Z_{j-1} + \theta \vartheta Z_{j-13}.$$

An $AR(1) \times AR(1)_{12}$ model would be

$$X_j = (\text{Id} - \phi B)^{-1}(\text{Id} - \varphi B^{12})^{-1} Z_j, \quad j \in \mathbb{Z}$$

and in a displayed form,

$$X_j - \varphi X_{j-12} - \phi X_{j-1} + \phi \varphi X_{j-13} = Z_j.$$

More general models are

$$\text{ARMA } (p, q) \times \text{ARMA } (a, c)_s,$$

or still more generally,

$$\text{SARIMA } (p, d, q) (a, D, c)_s.$$

The general equation of a SARIMA model is

$$\Phi_p(B) A_a(B^s)(\text{Id} - B)^d(\text{Id} - B^s)^D X_j = \mu + \Theta_q(B) C_c(B^s) Z_j$$

Applying this formula we can write some examples:

a) A SARIMA(0, 0, 0)(1, 0, 0)₁₂ is given by

$$X_j = \varphi X_{j-12} + W_j.$$

b) A SARIMA(0, 0, 0)(0, 1, 1)₄ is given by

$$X_j = X_{j-4} + W_j - \vartheta W_{j-4}.$$

c) A SARIMA(0, 1, 0)(1, 0, 0)₁₂ is given by

$$X_j = X_{j-1} + \varphi X_{j-12} - \varphi X_{j-13} + W_j.$$

d) A SARIMA(0, 1, 0)(0, 0, 1)₄ is given by

$$X_j = X_{j-1} + W_j - \vartheta W_{j-4}.$$

5 ARFIMA Models

It can be shown, thanks to Wold decomposition theorem, see [1] (page 76), that any second order stationary time series X can be written as

$$X_j = \sum_{i=0}^{\infty} \psi_i Z_{j-i} + f(t)$$

where f is a deterministic function. So, we assume in all the chapter that our process X is a centered purely nondeterministic stationary time series that has the representation

$$X_j = \sum_{i=0}^{\infty} \psi_i Z_{j-i}$$

with $\sum_{i=0}^{\infty} \psi_i^2 < \infty$ and Z is a centered white noise with variance σ^2 . Recall moreover, from Lecture 6, that these models include ARMA models and

$$\rho(l) = \frac{\sum_{r=0}^{\infty} \psi_r \psi_{r+l}}{\sum_{r=0}^{\infty} \psi_r^2}.$$

It is immediate to see that

$$\rho(l) \xrightarrow{l \uparrow \infty} 0.$$

Note that we can bound

$$\sum_{r=0}^{\infty} |\psi_r| |\psi_{r+l}| \leq \left(\sum_{r=0}^{\infty} |\psi_r|^2 \right)^{\frac{1}{2}} \left(\sum_{r=0}^{\infty} |\psi_{r+l}|^2 \right)^{\frac{1}{2}}$$

and the first term in the product is a finite constant and the second one can be re-written as

$$\left(\sum_{j=l}^{\infty} |\psi_j|^2 \right)^{\frac{1}{2}}$$

that of course converge to 0 when $l \uparrow \infty$ because $\sum_{r=0}^{\infty} |\psi_r|^2$ is finite.

Definition 5.1 *It is said that a stationary time series that satisfies*

$$\sum_{l=0}^{\infty} |\rho(l)| < \infty \tag{5.1}$$

has short memory. On the contrary, if property (5.1) is false, that is,

$$\sum_{l=0}^{\infty} |\rho(l)| = \infty. \tag{5.2}$$

we say the series has long memory.

ARMA processes have short memory; its autocorrelation function eventually decreases exponentially. In fact it can be shown that

$$|\rho(l)| \leq \frac{c}{|\alpha|^l}$$

where $c \geq 0$ and α is the smaller root of the AR polynomial, that of course, it is out of the unit circle, that is $|\alpha| > 1$. But not all stationary processes are short memory processes.

In Lecture 9 we saw that for second order causal linear models we have

$$\sum_{l=0}^{\infty} |\gamma(l)| \leq \sigma^2 \left(\sum_{r=0}^{\infty} |\psi_r| \right)^2$$

So, $\sum_{r=0}^{\infty} |\psi_r| < \infty$ is a sufficient condition to guarantee short memory. But recall that

$$\sum_{k=0}^{\infty} |\psi_k|^2 < \infty \tag{5.3}$$

is weaker than

$$\sum_{k=0}^{\infty} |\psi_k| < \infty \tag{5.4}$$

Then, there is room for stationary processes with long memory. There exists stationary second order causal linear models such that $\sum_{l=0}^{\infty} |\gamma(l)|$ and $\sum_{r=0}^{\infty} |\psi_r|$ are infinite.

Long memory processes appears in many problems in Finance, Hidrology or Meteorology. Examples of long memory processes are ARIMA (or FARIMA) processes, that we define below. They are stationary second order causal linear models, and so, they satisfy (5.3) but they have long memory, and so, they not satisfy (5.4).

Let Z be a centered white noise with variance σ^2 . Assume X satisfies

$$(Id - B)^d X = Z.$$

If $d = 0$, X is a white noise, or an ARIMA(0,0,0). If $d = 1$, X is a random walk, that is, an ARIMA(0,1,0). Can we consider a fractional $d \in (0,1)$? The answer is given by the following theorem, whose proof can be found in [2].

Theorem 5.2 *Let Z be a centered white noise with variance σ^2 . If $d \in (-\frac{1}{2}, \frac{1}{2})$, it exists a unique stationary solution of the equation*

$$(Id - B)^d X = Z$$

with

$$X_j = \sum_{i=0}^{\infty} \psi_i Z_{j-i},$$

and

$$\psi_i = \frac{\Gamma(i+d)}{\Gamma(i+1)\Gamma(d)} = \prod_{k=1}^i \frac{k-1-d}{k}.$$

In particular,

$$\gamma(0) = \frac{\Gamma(1-2d)\sigma^2}{\Gamma^2(1-d)},$$

$$\rho(l) = \frac{\Gamma(1-d)\Gamma(i+d)}{\Gamma(1+i-d)\Gamma(d)} = \prod_{k=1}^l \frac{k-1+d}{k-d}, \quad l \geq 1$$

and

$$\alpha(l) = \frac{d}{l-d}, \quad l \geq 1.$$

Recall that Gamma function for $x > 0$ is defined as

$$\Gamma(x) := \int_0^{\infty} t^{x-1} e^{-t} dt.$$

According to Stirling formula, for x big enough, we have

$$\Gamma(x) \sim \sqrt{2\pi} e^{1-x} (x-1)^{x-\frac{1}{2}}.$$

Using this formula we can determine accurately the speed of convergence of the coefficients of the causal series and the autocorrelations. If j is big enough, the coefficients behave as

$$\psi_j \sim \frac{1}{j^{1-d}\Gamma(d)}$$

Indeed,

$$\begin{aligned}
\frac{\Gamma(i+d)}{\Gamma(i+1)\Gamma(d)} &= \frac{e^{1-d}(i+d-1)^{i+d-\frac{1}{2}}}{\Gamma(d)i^{i+\frac{1}{2}}} \\
&= \frac{1}{\Gamma(d)}e^{1-d}\left(1+\frac{d-1}{i}\right)^i \frac{(i+d-1)^{d-\frac{1}{2}}}{i^{\frac{1}{2}}} \\
&= \frac{i^{d-1}}{\Gamma(d)}e^{1-d}\left(1+\frac{d-1}{i}\right)^i \frac{(i+d-1)^{d-\frac{1}{2}}}{i^{d-\frac{1}{2}}} \\
&= \frac{i^{d-1}}{\Gamma(d)}e^{1-d}\left(1+\frac{d-1}{i}\right)^i \left(1+\frac{d-1}{i}\right)^{d-\frac{1}{2}} \\
&\sim \frac{i^{d-1}}{\Gamma(d)}.
\end{aligned}$$

Similar estimations give

$$\rho(j) \sim \frac{\Gamma(1-d)}{\Gamma(d)} \frac{1}{j^{1-2d}}.$$

Indeed, from

$$\rho(j) = \frac{\Gamma(1-d)}{\Gamma(d)} \frac{\Gamma(j+d)}{\Gamma(1+j-d)}$$

we need only to check

$$\frac{\Gamma(j+d)}{\Gamma(1+j-d)} \sim \frac{1}{j^{1-2d}}.$$

We have

$$\begin{aligned}
\frac{\Gamma(j+d)}{\Gamma(1+j-d)} &= \frac{e^{1-2d}(j+d-1)^{j+d-\frac{1}{2}}}{(j-d)^{j-d+\frac{1}{2}}} \\
&= \frac{e^{1-2d}(j-d+2d-1)^{j-d+\frac{1}{2}+2d-1}}{(j-d)^{j-d+\frac{1}{2}}} \\
&= \frac{(j-d+2d-1)^{j-d+\frac{1}{2}} e^{1-2d}(j+d-1)^{2d-1}}{(j-d)^{j-d+\frac{1}{2}}} \\
&= \left(1+\frac{2d-1}{j-d}\right)^{j-d} e^{1-2d} (j+d-1)^{2d-1} \left(1+\frac{2d-1}{j-d}\right)^{\frac{1}{2}} \\
&= (j+d-1)^{2d-1} \left(1+\frac{2d-1}{j-d}\right)^{\frac{1}{2}} \\
&= \frac{(j+d-1)^{2d-\frac{1}{2}}}{(j-d)^{\frac{1}{2}}} \\
&= \frac{(j-d+2d-1)^{2d-\frac{1}{2}}}{(j-d)^{2d-\frac{1}{2}}(j-d)^{1-2d}} \\
&= \frac{1}{(j-d)^{1-2d}} \left(1+\frac{2d-1}{j-d}\right)^{2d-\frac{1}{2}} \\
&= \frac{1}{j^{1-2d}} \left(1-\frac{d}{j}\right)^{2d-1} \left(1+\frac{2d-1}{j-d}\right)^{2d-\frac{1}{2}}.
\end{aligned}$$

The proof finishes noticing that $(1 - \frac{d}{j})^{2d-1}$ and $(1 + \frac{2d-1}{j-d})^{2d-\frac{1}{2}}$ converge to 1.

Note that the series of ψ_j is divergent in absolute value for any $d \in [0, 1)$. If $d < 0$ the series is convergent and the model is of short memory.

In relation with coefficient $|\rho(l)|$, note that for $d \in [0, \frac{1}{2})$ the sequence $\rho(l)$ converge to 0 in absolute value but $1 - 2d \in (0, 1]$ and so

$$\sum_{l=0}^{\infty} |\rho(l)| = \infty.$$

Note that when d approaches $\frac{1}{2}$ the long memory effect increases. On the contrary, for $d < 0$ the model is a short memory one. If $d \geq \frac{1}{2}$ it can be shown the model is non stationary.

Definition 5.3 *A process X is an ARFIMA (p, d, q) model with $-\frac{1}{2} < d < \frac{1}{2}$ if and only if $(Id - B)^d X$ is an ARMA (p, q) . That is, if X is a stationary process that satisfies*

$$\Phi_p(B)(Id - B)^d X = \Theta_q(B)Z.$$

Note that we can write

$$\Phi_p(B)X = \Theta_q(B)(Id - B)^{-d}Z$$

so, we can interpret that an ARFIMA model is nothing more than an ARMA (p, q) model with a fractional noise given by $(Id - B)^{-d}Z$.

In relation with parameter estimation of ARFIMA models we have the following usual techniques

1. The mean is estimated by the empirical mean, but now this estimator is not asymptotically normal because the sum of the absolute values of the autocorrelations is not convergent.
2. The autocorrelation function is estimated by $\hat{\rho}$, but its asymptotic behaviour as estimator is not clear.
3. The d index and the parameters are estimated using least squares or maximum of likelihood.

References

- [1] P. J. Brockwell and R. A. Davis (1996): *Introduction to Time Series and Forecasting*. Springer.
- [2] W. Palma (2007): *Long memory time series, theory and methods*. Wiley.